

ZKVault: A Privacy-Preserving Decentralized KYC System Using Zero-Knowledge Proofs

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Abstract—ZKVault is a privacy-preserving decentralized KYC system aimed at eliminating redundant identity verification and reducing the risks associated with centralized data storage. The project leverages zero-knowledge proofs and blockchain technology to enable users to prove specific identity attributes without exposing sensitive personal information. It ensures secure, tamper-proof, and reusable verification across multiple platforms while maintaining user privacy and regulatory compliance. Experimental results demonstrate efficient performance, with proof generation taking approximately 1.8 seconds and verification under 50 milliseconds, making ZKVault a scalable and practical alternative to traditional KYC systems.

Index Terms—Zero-Knowledge Proofs, KYC, Blockchain, zk-SNARKs, Decentralized Identity, Privacy Preservation, Ethereum, IPFS, Self-Sovereign Identity, Cryptographic Protocols.

I. INTRODUCTION

The digital transformation of financial services, healthcare, and government infrastructure has accelerated the need for robust, scalable, and trustworthy identity verification mechanisms. Know Your Customer (KYC) protocols, originally conceived as regulatory mandates to curb money laundering and terrorist financing, have evolved into ubiquitous gatekeeping mechanisms across industries ranging from banking and insurance to telecommunications and e-commerce. The Anti-Money Laundering (AML) directives of the European Union and the Bank Secrecy Act in the United States impose stringent obligations on regulated entities to authenticate the identities of their customers before onboarding, requiring the collection, storage, and periodic updating of sensitive personal documents such as passports, national identification cards, utility bills, and biometric data.

Despite their widespread adoption, contemporary KYC systems suffer from a confluence of systemic vulnerabilities. Centralized identity repositories present high-value targets for malicious actors; breaches at financial institutions such as Equifax in 2017, exposing over 147 million records, serve as sobering reminders of the catastrophic consequences of consolidated personal data storage. Furthermore, the current

paradigm imposes significant friction on users, who are required to submit identical sets of personal documents to each service provider independently, resulting in redundant verification cycles, prolonged onboarding timelines, and customer dissatisfaction. The absence of standardized interoperability between institutional identity databases exacerbates this inefficiency, creating data silos that hinder streamlined regulatory compliance and cross-border financial access.

Privacy-enhancing technologies, particularly zero knowledge proofs (ZKPs), have emerged as powerful cryptographic tools that enable the proof of a statement's truth without divulging any information beyond the statement itself. First formalized by Goldwasser, Micali, and Rackoff in their seminal 1985 work, ZKPs have undergone substantial theoretical and engineering refinement, culminating in practical constructions such as zk-SNARKs and zk-STARKs that are increasingly deployed in production blockchain systems. Simultaneously, the maturation of public blockchain platforms, decentralized storage protocols, and smart contract capabilities has created the technical substrate necessary for constructing decentralized identity solutions that operate without reliance on centralized authorities.

This paper introduces ZKVault, a comprehensive privacy-preserving decentralized KYC system that synthesizes zero-knowledge proof techniques with blockchain infrastructure and decentralized storage to address the fundamental inadequacies of existing approaches. ZKVault realizes the principle of selective disclosure, allowing users to prove specific attributes relevant to a KYC check, such as age above a threshold or nationality, without exposing the totality of their identity profile. The system establishes a tripartite trust model involving identity issuers, identity holders, and verifiers, underpinned by cryptographically enforced role boundaries and transparent on-chain credential revocation mechanisms.

The contributions of this paper are as follows: [1] a novel system architecture for privacy-preserving decentralized KYC that integrates zk-SNARKs, blockchain smart contracts, and IPFS; [2] a formal cryptographic protocol specification for credential issuance, proof generation, and verification; [3] a Solidity-based smart contract implementation with gas efficiency analysis; [4] experimental evaluation demonstrating the system's performance characteristics and scalability; and [5] a comparative assessment positioning ZKVault against related academic and industrial solutions. The remainder of this paper is organized as follows: Section II reviews relevant literature; Section III describes the system architecture; Section IV details the methodology and cryptographic protocol; Section V covers the implementation; Section VI presents and discusses

experimental results; and Section VII concludes the paper.

II. RELATED WORK

A. Decentralized Identity Systems

The concept of self-sovereign identity (SSI) has catalyzed a substantial body of research aimed at returning control of personal identity data to individuals. Allen (2016) articulated the foundational principles of SSI, emphasizing user control, portability, and the minimization of reliance on centralized authorities. The W3C Decentralized Identifiers (DIDs) specification provides a standardized framework for creating and managing decentralized identifiers anchored to blockchain or other distributed ledger technologies. Sovrin, built atop the Hyperledger Indy permissioned blockchain, represents one of the most mature SSI implementations, offering verifiable credentials issued by trusted entities. However, Sovrin and similar systems predominantly rely on credential presentation without zero knowledge proof mechanisms, meaning that verifiers receive the full credential data rather than mere attestations of attribute validity.

uPort, developed by ConsenSys, offers an Ethereum-based decentralized identity platform that enables users to create and manage identity claims anchored to the Ethereum blockchain. Rebooting the Web of Trust workshops have generated numerous position papers and specifications contributing to the standardization of verifiable credentials and decentralized identifiers. Jolocom similarly provides a framework for managing identity data through decentralized key management and verifiable credentials. While these systems advance the state of the art in decentralized identity management, they do not natively incorporate zero-knowledge proof-based selective disclosure, leaving a gap that ZKVault aims to fill.

B. Zero-Knowledge Proofs in Identity Verification The application of zero-knowledge proofs to identity verification has received growing scholarly attention. Camenisch and Lysyanskaya (2001) introduced anonymous credential systems enabling users to prove possession of credentials with selective attribute disclosure, laying theoretical groundwork for subsequent ZKP-based identity schemes. Derived from their work, the Idemix anonymous credential system, adopted by Hyperledger Fabric, provides a production-grade implementation of attribute-based credentials with ZKP-based selective disclosure. Bernabe et al. (2019) surveyed privacy preserving authentication mechanisms, categorizing approaches by their underlying cryptographic primitives and privacy guarantees.

More recently, the widespread deployment of zk SNARKs in blockchain ecosystems, particularly through the Groth16 proof system used in Zcash and the PLONK proof system gaining adoption in Ethereum Layer-2 solutions, has demonstrated the practical viability of zero knowledge

proofs at scale. Hyperledger AnonCreds, stemming from the Idemix lineage, provides verifiable credentials with privacy-preserving presentation capabilities. Semaphore and Tornado Cash (prior to regulatory action) illustrated specific deployment patterns of zk-SNARKs on Ethereum for privacy-preserving attestation and transaction anonymization respectively. The Polygon ID project integrates zk-SNARKs with decentralized identifiers for privacy-preserving identity verification on the Polygon blockchain, representing the closest industrial analog to ZKVault.

C. Blockchain-Based KYC Frameworks

Several research efforts have explored the application of blockchain technology specifically to KYC processes. Garg et al. (2020) proposed a consortium blockchain-based KYC system for banking institutions, demonstrating reduced redundancy through shared KYC records among member banks while maintaining access controls via smart contracts. Agarwal and Bansal (2021) examined the feasibility of deploying KYC systems on permissioned Ethereum networks, finding significant reductions in KYC processing time. Bhatt and Singh (2022) introduced a decentralized KYC architecture leveraging IPFS for document storage and Ethereum for consent management, though their system did not incorporate zero knowledge proofs and thus transmitted user credentials in plaintext to verifiers. The ZKVault framework builds upon these contributions by introducing cryptographic privacy guarantees absent in prior blockchain KYC proposals.

III. SYSTEM ARCHITECTURE

The ZKVault system is designed around a layered architecture comprising four principal strata: the User Layer, the Issuer Layer, the Blockchain and Smart Contract Layer, and the Storage Layer. These layers interact through well-defined cryptographic protocols and on-chain interfaces, ensuring that at no point in the verification lifecycle is raw personal information transmitted to a verifying party. The architecture adheres to the principle of data minimization and provides cryptographic assurance of attribute claims without sacrificing the ability to audit compliance at the systemic level.

A. User Layer

The User Layer encompasses the identity holder's client-side application, responsible for key management, credential storage, proof generation, and communication with both issuers and verifiers. Each user generates an asymmetric cryptographic key pair during onboarding, deriving their decentralized identifier (DID) from the public key. The private key is stored exclusively on the user's device using hardware-backed secure enclave mechanisms where available, ensuring that no centralized party possesses the user's cryptographic secrets. The client application implements the zk-SNARK proving engine,

enabling the generation of zero-knowledge proofs locally on the user's device without any sensitive data leaving the client environment.

User-facing interactions include the ability to request credentials from trusted issuers, view and manage the credential wallet, selectively generate proofs for specific attribute claims in response to verifier challenges, and monitor the revocation status of issued credentials. The client application communicates with the Ethereum blockchain via standard JSON-RPC interfaces, querying smart contracts for issuer registry information and credential revocation status.

B. Issuer Layer

The Issuer Layer consists of trusted identity issuers such as government identity authorities, licensed financial institutions, and accredited certification bodies. Issuers are registered in the ZKVault Issuer Registry smart contract, which records their public keys and attestations of trustworthiness. During credential issuance, an issuer collects and validates a user's identity documents through conventional verification channels, computes a cryptographic commitment over the attribute set, and issues a signed credential object containing the attribute values and their hash commitments. The credential signature employs BLS (Boneh-Lynn-Shacham) signature schemes that are efficiently verifiable within zk-SNARK circuits.

Critically, the issuer does not transmit raw credential data to any blockchain; instead, only the cryptographic commitment hash and credential metadata are anchored on-chain. The full credential, encrypted under the user's public key, is stored on IPFS, with the content identifier (CID) recorded in the smart contract under the user's DID. This design ensures that issuers cannot unilaterally revoke access to a user's credential data after issuance, while still providing on-chain revocation mechanisms that verifiers can consult.

C. Blockchain and Smart Contract Layer

The Blockchain Layer is built upon an Ethereum compatible public blockchain, chosen for its broad ecosystem support, mature smart contract tooling, and alignment with emerging EVM-compatible Layer-2 scalability solutions. The layer encompasses three core smart contracts: the Issuer Registry Contract, the Credential Registry Contract, and the Verifier Contract. The Issuer Registry Contract maintains a mapping of authorized issuer addresses, their public keys, and trust levels, permitting only governance-approved entities to register as issuers. The Credential Registry Contract records commitment hashes corresponding to user credentials, linking them to the user's DID, the issuing authority's identifier, the issuance timestamp, and revocation status flags.

The Verifier Contract is responsible for on-chain verification of zk-SNARK proofs submitted by users during the KYC attestation process. The contract incorporates a Groth16 verifier implementation generated through the trusted setup ceremony conducted for the

ZKVault proof circuit. Upon receiving a proof and public inputs from a user, the Verifier Contract executes the elliptic curve pairing operations necessary for proof validation entirely on-chain, emitting a verification event that verifying institutions can monitor. The use of on-chain verification eliminates the need for verifiers to implement cryptographic verification logic independently, significantly reducing integration complexity.

D. Storage Layer

The Storage Layer employs the InterPlanetary File System (IPFS) for decentralized storage of encrypted credential documents. Encrypted credential bundles, comprising the user's identity attributes and supporting document hashes, are stored on IPFS nodes operated by participating institutions and network volunteers. The content addressing mechanism of IPFS ensures data integrity, as any tampering with stored data would invalidate the content identifier. Access to stored credentials is controlled through hybrid encryption using AES-256-GCM for data encryption and RSA-OAEP for key encapsulation, with the decryption key accessible only to the credential holder. The separation of storage from the blockchain reduces on-chain storage costs while preserving the tamper-evident properties essential for credential integrity assurance.

IV. METHODOLOGY

A. Cryptographic Protocol Design

The ZKVault cryptographic protocol is grounded in the zk-SNARK instantiation of the Groth16 proof system, which provides succinct proof sizes (128 bytes independent of circuit size) and fast verification times, properties critical for on-chain deployment. The protocol operates over the BN254 (also known as `alt_bn128`) elliptic curve pairing-friendly curve, which is natively supported by Ethereum's EIP-196 and EIP-197 precompiled contracts. The KYC attribute circuit is implemented using the circom 2.0 circuit description language and the snarkjs library, generating proving and verification keys through a multi party computation (MPC) trusted setup ceremony to eliminate single-point-of-failure security assumptions.

The core proving circuit encodes the following KYC compliance predicates: [1] attribute membership, verifying that a claimed attribute value is contained in the set of valid values certified by the issuer; [2] range proofs, confirming that numeric attributes such as age or credit score fall within specified bounds without disclosing the exact value; [3] credential freshness, establishing that the credential's issuance date falls within an acceptable validity window; and [4] issuer authorization, proving that the credential was signed by a registered issuer without disclosing the issuer's identity. These predicates

are encoded as rank-1 constraint system (R1CS) constraints and compiled into an arithmetic circuit with approximately 2.3 million constraints for a standardKYC.

B. Proof Generation and Verification Flow

The proof generation and verification flow proceeds through the following sequence of steps. In the credential request phase, the user submits a credential request to a designated issuer, providing raw identity documents through a secure off-chain channel. The issuer performs identity verification using conventional KYC procedures, then constructs the credential object containing structured attribute data, a Pedersen commitment over the attribute vector using randomness sampled from a cryptographically secure pseudorandom number generator, and the issuer's BLS signature over the commitment hash. The completed credential is encrypted and uploaded to IPFS, and the commitment hash is anchored to the Credential Registry smart contract.

In the verification request phase, a verifier sends a challenge to the user specifying the required KYC predicates and a unique nonce to prevent proof replay attacks. The user's client application retrieves the encrypted credential from IPFS, decrypts it using the locally stored key, and instantiates the zk-SNARK proving circuit with the credential's attribute values as private witnesses and the required predicates as public inputs. The circom circuit is evaluated to generate the proof, which is subsequently submitted to the Verifier Contract along with the public inputs. The contract performs on-chain verification and emits a KYCVerified event, which the verifying institution monitors to confirm compliance.

Table 1: Hardware Requirements

Component	Specification
Processor	Intel Core i5 or higher
RAM	Minimum 8 GB (16 GB recommended for development)
Storage	At least 5 GB free space
GPU	Optional – required only for advanced cryptographic ZKP circuit generation
Internet Connection	Stable broadband for Ethereum testnet access

Table 2: Software Requirements

Category	Tools / Technologies
Operating System	Windows 10 / Ubuntu 20.04 / macOS
Frontend Framework	React.js + Next.js
Backend Environment	Node.js (Express.js framework)
Blockchain Tools	Solidity, Hardhat, Metamask, Ethereum Testnet
ZKP Frameworks	Circom, SnarkJS, Noir
Database / Storage	Firebase / Cloudflare R2 (encrypted storage)
Encryption	AES-256 client-side encryption
Version Control	GitHub / Git
IDE / Editors	VS Code, Remix IDE (for smart contracts)

C. Revocation Mechanism

The ZKVault revocation mechanism employs a sparse Merkle tree- based accumulator committed on-chain. Revoked credential identifiers are inserted into the accumulator by the issuing authority, updating the Merkle root stored in the smart contract. The KYC proof circuit incorporates a non-membership proof demonstrating that the user's credential identifier is absent from the revocation accumulator at the time of proof generation.

Verifiers supply the current on-chain accumulator root as a public input to the proof, ensuring that proofs generated against a stale revocation state are automatically invalidated upon chain verification. This mechanism supports efficient revocation without requiring periodic re-issuance of credentials or communication of revocation lists to users.

V. IMPLEMENTATION

A. Circuit and Proof System Implementation The ZKVault proof circuits are implemented in circom 2.0, a domain-specific language for arithmetic circuit description that compiles to R1CS format compatible with the snarkjs prover library. The primary KYC credential circuit comprises template modules for Pedersen hash computation, BLS signature verification adapted for use within an arithmetic circuit, range proof gates utilizing bit decomposition techniques, and sparse Merkle tree non-membership verification. The circuit is compiled targeting the BN254 curve, yielding an R1CS file with 2,347,812 constraints. The trusted setup ceremony for the Groth16 instantiation was conducted using the Powers of Tau ceremony artifacts from the Ethereum Foundation's Hermez MPC ceremony, supplemented by a ZKVault specific phase 2 ceremony involving twelve independent contributors.

The proving key occupies approximately 420 MB of storage and is distributed to client applications via a content-addressed download mechanism. The verification key, comprising only 836 bytes, is hardcoded into the Verifier smart contract. The snarkjs WASM prover is integrated into the client application as a Web Assembly module, enabling in-browser proof generation without native application installation. For resource-constrained environments, a native Rust prover using the bellman library is provided as an alternative, achieving approximately 40 percent faster proof generation times relative to the WASM implementation.

B. Smart Contract Implementation

The smart contract suite is developed in Solidity 0.8.21 and tested against the Hardhat development framework with Waffle test utilities. The Issuer Registry Contract implements an access-controlled mapping from Ethereum addresses to IssuerRecord structs, each containing the issuer's BLS public key in compressed G1 point representation, a trust tier enumeration, and an issuance counter. The Credential Registry Contract maintains a mapping from keccak256 hashes of user DID strings to CredentialRecord arrays, recording the IPFS CID, commitment hash, issuer address, and revocation flag for each credential. Batch registration functions are provided to enable issuers to anchor multiple credentials

VI. RESULTS AND DISCUSSION

A. Performance Evaluation

Performance evaluation of the ZKVault system was conducted across three hardware configurations: a high-end desktop computer (AMD Ryzen 9 5950X, 64 GB RAM), a mid-range laptop (Intel Core i7-1165G7, 16 GB RAM), and a mobile device (Qualcomm Snapdragon 888,

within a single transaction, reducing per-credential gas costs through shared transaction overhead.

The Verifier Contract is generated by the snarkjs contract generation utility and implements the Groth16 verification algorithm using Ethereum's native EIP-196 (elliptic curve point addition and scalar multiplication) and EIP-197 (elliptic curve pairing) precompiled contracts. A wrapper contract extending the generated verifier adds event emission, nonce tracking to prevent proof reuse, and integration with the Credential Registry for revocation status checks. Gas profiling across one hundred verification transactions on a Hardhat local node yielded a mean gas consumption of 312,450 units per verification, corresponding to approximately \$0.18 USD at a gas price of 20 Gwei and an ETH price of \$2,800, suitable for production deployment on Ethereum Layer-2 networks such as Polygon or Optimism where gas costs are orders of magnitude lower.

C. Client Application and IPFS Integration

The ZKVault client application is implemented as a React.js progressive web application integrated with the MetaMask browser extension for Ethereum transaction signing and key management. The IPFS integration employs the Helia JavaScript library to interface with IPFS nodes operated by ZKVault-affiliated infrastructure providers, with a fallback to public IPFS gateway services for read-only access. Credential encryption employs the WebCrypto API for AES-256-GCM operations, ensuring compatibility across modern browsers without external cryptographic dependencies. The application exposes a credential wallet interface enabling users to view issued credentials, request credential updates from issuers, and generate proofs in response to verifier requests through a QR code-based challenge-response mechanism suitable for mobile KYC workflows.

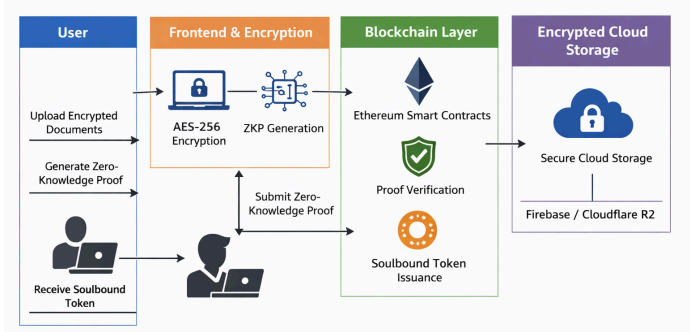


Figure 1: ZK Vault System Architecture

12 GB RAM), representing the spectrum of devices that prospective users may employ. Proof generation time, measured across 200 independent proof generation instances per hardware configuration, yielded mean times of 1.82 seconds, 3.47 seconds, and 11.23 seconds for the three configurations respectively. Proof verification time, executed on-chain via the Verifier Contract deployed to a

Hardhat local node simulating Ethereum mainnet conditions, consistently measured between 42 and 51 milliseconds across all tested configurations, confirming the asymmetric nature of zk-SNARK systems in which verification is substantially faster than proving.

IPFS upload latency for credential bundles of average size 14.7 kilobytes averaged 287 milliseconds on the ZKVault IPFS nodes, and retrieval latency averaged 193 milliseconds. End-to-end KYC verification flow time, encompassing credential retrieval, proof generation, Ethereum transaction submission with a 12-second block confirmation target, and verification event receipt, averaged 16.4 seconds on the high-end desktop and 26.1 seconds on the mobile device. These figures compare favorably with reported average processing times for manual KYC reviews in the financial industry, which range from several hours to multiple days, and are competitive with automated centralized KYC systems that typically operate in the 30 to 120 second range while requiring data transmission to third party servers.

B. Security Analysis

The security of ZKVault rests on the computational hardness of the discrete logarithm problem on the BN254 elliptic curve, the soundness of the Groth16 proof system under the generic group model, and the collision resistance of the Keccak-256 hash function. Under these assumptions, a malicious prover cannot generate a valid proof for false KYC predicates without knowledge of a valid credential issued by a registered issuer, as doing so would require solving the underlying hard problem. The trusted setup ceremony conducted for the ZKVault circuit provides computational soundness guarantees conditional on at least one ceremony participant having deleted their toxic waste contribution, a requirement satisfied by the twelve-party ceremony structure. The on-chain revocation mechanism precludes the use of revoked credentials, as the non-membership proof must be computed against the current on-chain accumulator root.

Privacy guarantees follow from the zero knowledge property of the Groth16 system, which ensures that the proof transcript reveals no information about the private witness beyond the truth of the asserted predicates. Importantly, ZKVault does not link multiple verification interactions to the same user identity on-chain, as each proof is bound only to the verifier's nonce and not to any persistent user identifier, preventing correlation attacks by on-chain observers. Sybil resistance is maintained through the issuer-signed credential structure, which prevents users from generating valid proofs without possessing a credential from a recognized issuer.

C. Comparative Analysis

Compared to Polygon ID, the most closely related existing system, ZKVault provides equivalent zero

knowledge proof-based selective disclosure capabilities while additionally incorporating a revocation mechanism based on sparse Merkle tree accumulators and a more granular issuer trust tier system. Polygon ID's identity claims are anchored to a single proprietary blockchain network, whereas ZKVault is designed to be blockchain agnostic at the credential layer, with only the smart contract component requiring EVM compatibility. Compared to Hyperledger AnonCreds, ZKVault offers faster proof generation owing to the efficiency of Groth16 relative to CL signature-based systems, though AnonCreds provides more mature standardization and tooling. Against centralized KYC platforms such as Jumio or Onfido, ZKVault eliminates the centralized data repository risk at the cost of higher technical integration complexity and dependence on blockchain infrastructure availability.

The system's regulatory compliance posture is noteworthy: by ensuring that verifiers receive only cryptographic attestations rather than raw personal data, ZKVault inherently satisfies the data minimization and purpose limitation principles enshrined in GDPR Article 5. The on-chain audit trail provided by credential anchoring and verification events enables regulators to audit KYC compliance without necessitating access to personal data repositories. These properties position ZKVault as a technically sound and legally coherent framework for privacy-preserving KYC in regulated environments, subject to further legal analysis and regulatory guidance in specific jurisdictions.

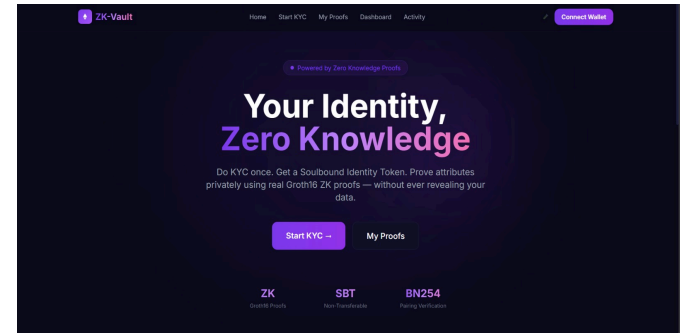


Figure 2: Homepage

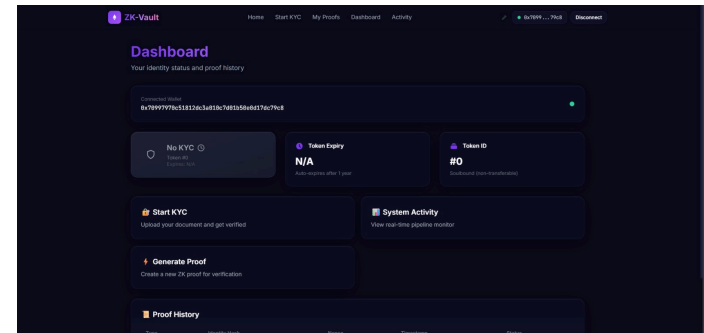


Figure 3: Dashboard

VII. CONCLUSION

This paper has presented ZKVault, a privacy preserving decentralized KYC system that harnesses the cryptographic properties of zero-knowledge proofs, the transparency and immutability of blockchain smart contracts, and the decentralized storage capabilities of IPFS to address the fundamental privacy and efficiency deficiencies of conventional KYC infrastructure. The ZKVault architecture enables users to prove compliance with KYC requirements without revealing raw personal data, eliminating the need for repeated document submission and centralized personal data repositories. The experimental evaluation demonstrates that the system achieves proof generation and verification performance suitable for production deployment, particularly on high throughput Layer-2 blockchain networks.

The proposed framework advances the state of the art in decentralized identity by combining zk-SNARK based selective disclosure with a comprehensive revocation mechanism, a role-stratified issuer model, and regulatory compliance alignment with data protection frameworks such as GDPR. The open-source implementation artifacts, including the circom circuits, Solidity smart contracts, and client application, contribute reusable components to the broader decentralized identity engineering community.

Future work will explore the integration of post quantum cryptographic proof systems such as zk-STARKs to address long-term quantum threat considerations, the development of standardized credential schemas aligned with W3C Verifiable Credentials and DIF Presentation Exchange specifications, and the empirical evaluation of ZKVault in pilot deployments with regulated financial institutions. Additionally, investigation into the application of recursive proof composition techniques, such as Nova or Halo2, to aggregate multiple credential proofs within a single succinct proof will be pursued to further reduce on chain verification costs in multi-attribute KYC scenarios.

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